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CLINICAL GUIDANCE

PM&R

Multi-disciplinary collaborative consensus guidance statement on the assessment and treatment of breathing discomfort and respiratory sequelae in patients with post-acute sequelae of SARS-CoV-2 infection (PASC)

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INTRODUCTION

Patients infected with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), the virus responsible for coronavirus disease 2019 (COVID-19), may experience lasting sequelae following the acute phase of their index infection.[1] The post-acute sequelae of SARS-CoV-2 infection (PASC) can manifest as a wide-ranging constellation of disabling symptoms. These symptoms are sometimes referred to as being part of long COVID, post-acute COVID-19 syndrome, long-haul COVID, chronic COVID, or other terms, but for the purposes of this report, the term PASC will be used. Respiratory symptoms are among the most common symptoms reported by patients, and include shortness of breath, impaired exercise tolerance, cough, and chest pain.[2-7]

Although the incidence of PASC is not yet fully known, estimates suggest varying length of symptoms post-initial infection and expected recovery. In a study of patients with incident cases of COVID-19 in which individuals self-reported their symptoms prospectively in the COVID Symptom Study, a total of 558 participants (13.3%) reported symptoms lasting ≥28 days, 189 (4.5%) for ≥8 weeks, and 95 (2.3%) for ≥12 weeks.[8] Persistent respiratory symptoms, in particular, may be more common among patients who required hospitalization, as dyspnea was present in 30% of previously hospitalized patients at 12-month follow-up in one longitudinal cohort.[9]

Despite the prevalence of these sequelae and emerging data on the longevity of symptoms, limited guidance exists regarding the assessment and treatment of PASC. The American Academy of Physical Medicine and Rehabilitation (AAPM&R) Multi-Disciplinary PASC Collaborative ("PASC Collaborative") was convened to address the pressing need for guidance in the care of patients with PASC. This document is part of a larger series addressing PASC, and specifically addresses the assessment and treatment of breathing discomfort and respiratory sequelae.

We acknowledge that the definition of PASC is evolving, and there are various factors that contribute to the diagnosis. The PASC Collaborative sought input from patient representatives and patient-led research initiatives to inform recommendations. Literature available at the time of our consensus process suggested that PASC be defined as the persistence of symptoms beyond 3 or 4 weeks from the onset of acute infection.[1] Alternative definitions of PASC include symptoms lasting longer than 3 months.[10] Following the completion of our consensus process for this report, World Health Organization (WHO) released a definition of "post-COVID condition," including describing the timing as "usually 3 months from the onset of COVID-19" and lasting "for at least 2 months."[11] Based on patient feedback during our consensus process, we believe that earlier evaluation, diagnosis, and management can improve access to beneficial interventions. For the purposes of this guidance statement, we recommend expanded assessment if symptoms are not improving 1 month after acute symptom onset.

PASC consensus guidance statement methods

The PASC Collaborative was created, in part, to develop expert recommendations and guidance from established PASC centers with extensive experience in managing patients with PASC. The PASC Collaborative is following an iterative modified Delphi approach to achieve consensus on assessment and treatment recommendations for a series of Consensus Guidance Statements focused on the most prominent PASC symptoms.[12,13] The collaborative is composed of

TABLE 1 PASC breathing discomfort assessment recommendations

#	Statement
1a	Clinicians should conduct a full patient history including review of predisposing comorbidities, characterizing course of acute COVID-19 (eg, medications received, oxygen saturation during acute illness, need for intensive care unit admission and mechanical ventilation), use of supplemental oxygen during or after acute illness, activity level since COVID-19, medications, and any new diagnoses or complications acquired following the index illness.
1b	Clinicians should systematically characterize breathing discomfort, including use of standard measures to describe and quantify the discomfort (see Discussion), and understand activity level that patients can achieve. The contribution of other factors that limit activity, including fatigue with post-exertional malaise and impaired hemodynamic response to activity (eg, inappropriate sinus tachycardia), must be considered in this activity assessment.
1c	Clinicians should assess the trajectory of breathing discomfort over time (ie, improving, worsening, or unchanged) to triage need for further workup.
2	Obtain pre-COVID-19 history including baseline respiratory symptoms and exercise capacity; obtain and review prior and recent pulmonary testing, including pulmonary function tests, chest radiograph, and chest computed tomography scan, when available.
3	In addition to full vital signs and physical exam, assess pulse oximetry while walking in a clinic office, breathing ambient air, to assess for desaturation. The pace and/or duration of exertion during this test can be increased, as tolerated and feasible.
4	Consider performing pulmonary function tests (PFTs) in patients with persistent breathing discomfort that is not improving at least 8 weeks after acute COVID-19, or in patients who develop new onset or worsening breathing discomfort later in the post-COVID course. Consider consultation with a pulmonologist for new or progressive abnormalities.
5	Chest imaging should be considered in evaluating breathing discomfort in the setting of ambulatory desaturation, abnormal pulmonary exam, impairments on pulmonary function testing, or on an individual basis for other clinical concerns. Chest radiographs are an appropriate initial study for most patients, although additional considerations are discussed within this document. Consider consultation with a pulmonologist or PASC clinic to guide imaging.
6	Transthoracic echocardiography, cardiac stress testing, and cardiopulmonary exercise testing are not routinely recommended to evaluate breathing discomfort alone after COVID-19. However, these tests may be valuable to consider in the context of a patient's associated symptoms (eg, chest pain) and treatment plan (eg, rehabilitation therapy) on an individual basis in consultation with a PASC clinic, cardiologist, pulmonologist, or pulmonary rehabilitation physiatrist.
7	Consider evaluation by an otolaryngologist for abnormal upper airway breath sounds or voice changes, including stridor, hoarseness, or unexplained episodic breathing discomfort, especially in patients with a history of endotracheal intubation and/or tracheostomy following critical illness related to COVID-19.
8	For patients undergoing rehabilitation therapy, obtain standardized measures of activity performance (see Discussion). For patients physically limited by fatigue with post-exertional malaise, neurological impairment, or other impairments, perform alternative measures of activity performance (see Discussion and refer to the Consensus Statement on Fatigue in PASC Patients).[15]

TABLE 2 PASC breathing discomfort treatment recommendations

#	Statement
1	Consider evaluation by a pulmonologist for any of the following, based on assessment recommendations: • abnormal PFTs and/or chest imaging; • abnormal pulmonary exam, which is persistent, unexplained, and/or unresolved after management in primary care setting; • persistent productive cough and/or difficulty clearing airway secretions; and • oxygen desaturation with activity.Referral to a pulmonologist for breathing discomfort in the absence of these abnormalities can be considered on an individual basis when persistent or unexplained.
2	Patients requiring home oxygen (see Discussion) should be provided a portable oxygen device whenever possible to maximize mobility, ability to participate in rehabilitation, and quality of life. Oxygen should be appropriately titrated to an ambulatory saturation range based on established standards, to facilitate progress with rehabilitation and activity.
3	Refer patients with persistent breathing discomfort resulting in activity limitation for individualized rehabilitation therapy (Figure 1): a. If patients have accompanying fatigue with post-exertional malaise and/or dysautonomia, the most physically limiting factor should dictate the pace of activity as these can be worsened by over-activity.[15] b. If patients are awaiting PFTs, they can undergo general rehabilitation therapy while waiting if they are not thought to require further cardiac clearance prior to exercise. c. Consider physiological and subjective response to activity to determine the appropriate paced approach, similar to other aspects of PASC including fatigue and dysautonomia. d. Patients with PFT abnormalities meeting criteria for pulmonary rehabilitation, with associated symptoms and functional limitations, should be referred for pulmonary rehabilitation, whenever possible (see Discussion).
4	Provide breathing exercises through self-directed educational resources (see Discussion), in-person rehabilitation, or online programs.
5	For patients with breathing discomfort that is slowly improving after COVID-19, or for those without access to supervised rehabilitation, provide information (see Discussion) on self-monitored paced physical activity and breathing therapies. For patients with phone-based or wearable activity trackers, use data to track progress of therapy.
6	Instruct patients with chronic productive cough, difficulty clearing airway secretions, or with existing or new evidence of bronchiectasis regarding airway clearance techniques and consider prescribing an airway clearance device, where appropriate (see Discussion).
7	Pharmacologic therapies, including oral corticosteroids, inhaled bronchodilators, and inhaled corticosteroids, are not routinely recommended for breathing discomfort in the absence of impaired pulmonary function. Treatments may be considered when supported by objective findings (eg, examination, diagnostics). Consultation with a pulmonologist may be considered to assist this decision.

PFT, pulmonary function tests.

34 established post-COVID-19 or PASC centers, the first of which were established in April to June 2020; over 50 experts spanning clinical disciplines and specialties participate in the monthly meetings. Areas of expertise included physical medicine and rehabilitation, pulmonology, cardiology, primary care, physical therapy, speech therapy, and respiratory therapy, among others. Individual guidance statements include acknowledgement of the total number of centers participating in the Collaborative at the time of the publication, given the evolving nature of this group. Participating PASC centers were asked to designate one expert to be the voting member to assess consensus. Patient representatives were also permitted to vote. These statements were developed by a diverse team of experts, with patient input, and integrate current experience and expertise with available evidence to provide tools to clinicians treating patients. There is an intentional focus on health equity because disparities in care and outcomes are critically important to address. Beyond offering recommendations for assessment and treatment based on experience with care of patients presenting with PASC symptoms, the hope is that a broadened understanding of current patient care practices will help identify areas of future research. As with other PASC Collaborative documents, a detailed literature review was performed prior to initiation of the modified Delphi approach, the full description of our methodology has been published in detail previously.[14]

Considerations and caveats for implementation

These Guidance Statements are intended to reflect current practice in patient assessment, testing, and treatments. At the time of development, must of the literature focused on patients who were not vaccinated, and the incidence and trajectory of PASC in vaccinated patients with "breakthrough" cases (including but not limited to current and emerging variants of the virus) is evolving. The PASC Collaborative took this into account during the development process, and these guidance statements generally apply to individuals who develop PASC regardless of their vaccination status.

It is important to note that the recommendations provided in the Guidance Statements should not preclude clinical judgment and must be applied in the context of the specific patient, with adjustments for patient preferences, comorbidities, and other factors. As with any treatment plan, clinicians treating patients with PASC are encouraged to discuss the unknowns of PASC treatments and prognosis, as well as the benefits and risks of any treatment approach. This document specifically addresses post-acute persistent breathing discomfort and respiratory sequelae in the outpatient setting and does not apply to the acute assessment or management of breathing discomfort in urgent or emergency care settings.

In addition, the PASC Collaborative recognizes that patients with PASC typically present with a cluster of symptoms that cross multiple body systems, such as fatigue and dysautonomia, which might limit their ability to participate fully in some of these assessment and treatment recommendations. These co-existing conditions must always be considered when suggesting a test or treatment, such as physical rehabilitation. Assessment and treatment of these co-existing conditions has[15] and will be addressed in additional AAPM&R consensus statements.

Respiratory symptoms, signs, and testing abnormalities

Breathing discomfort and respiratory sequelae of COVID-19 may include disabling symptoms, abnormal clinical exam findings, and abnormalities on cardiopulmonary testing or chest imaging studies.[16-18] Common symptoms include shortness of breath at rest; disproportionate shortness of breath with activity; chest discomfort that may manifest as pain, tightness, constriction, and/or pressure; inability to take a full deep breath; new or progressive cough; and chest congestion.[17-19] These symptoms frequently result in limited activity tolerance.[16,19,20]

Clinical examination may be normal or reveal abnormalities including tachypnea, tachycardia, hypoxemia at rest or with exertion, wheezing, rales, or abnormalities on cardiac exam. On further evaluation, some patients may have evidence of lung disease, such as abnormal pulmonary function tests (PFTs) and/or abnormalities on chest imaging, inclusive of but not limited to chest radiographs and computed tomography (CT) of the chest.[16-18]

The severity of lung disease and abnormalities in pulmonary function in individuals with PASC appears to be associated with severity of the acute COVID-19 episode. It has been established previously that patients surviving acute respiratory distress syndrome (ARDS) can have persistent abnormalities in pulmonary function.[21] Similarly, 4 or more months after severe, or less commonly, moderate COVID-19 infection, changes in pulmonary function and chest imaging have been described, although data on the time course of this process are limited.[22-25] The most common abnormality described is impaired diffusion capacity on pulmonary function testing—found in $\sim$40% of patients at 1-year follow up after hospitalization for COVID-19 in China.[26] A systematic review of studies examining pulmonary function abnormalities after acute COVID-19 similarly found diffusion capacity impairments in 39% of patients across seven studies.[20] Six months after hospitalization for severe COVID-19, one cohort study identified changes concerning for pulmonary fibrosis in approximately one-third of patients—these changes were most common in older patients and those who experienced ARDS.[27] At this time, the time course of pulmonary fibrosis identified in individuals with PASC is unknown and it is unclear whether the condition will persist, progress, improve, or resolve.

Patients who initially had mild COVID-19, and did not experience hypoxemia or require hospitalization, are less likely to have post-acute pulmonary function or imaging abnormalities.[22] However, shortness of breath and breathing discomfort remain common aspects of PASC among patients with initial mild acute COVID-19 and warrant close evaluation, as described in this report.[28]

As noted in the AAPM&R Multi-Disciplinary PASC Consensus Guidance Statement methodology,[14] the recommendations (Tables 1 and 2) are based on expert consensus. Specific guidance recommendations that have been approved by consensus will be noted in the recommendation tables and are accompanied by additional discussion when appropriate.

Assessment recommendations discussion

History and physical

During the initial evaluation of breathing discomfort in the post-acute period, health care providers should determine whether lung disease is present and whether pulmonology evaluation is warranted. The severity of acute COVID-19 may help to initially stratify the risk of lung disease. Thus, history should include a detailed accounting of a patient's acute illness and the trajectory of symptoms since that time, in addition to history of pulmonary or cardiac disease. Characterizing the sensations and severity of breathing discomfort may also help guide further testing and treatment, as discussed in the section that follows. As noted previously, ARDS survivors are more likely to have impairments in pulmonary function testing and persistent interstitial lung abnormalities, when compared to those with less severe acute illness.[21,22] However, when estimating initial severity, clinicians must also account for resource limitations and inequities in care that may confound the characterization of the acute illness—for instance, patients who remained at home and were not hospitalized may have had hypoxemia that was not identified and treated in the acute phase, or may not have had access to inpatient hospitalization or home pulse oximetry (refer to Table 3: Health Equity Considerations and Examples in Patients with PASC: Breathing Discomfort). Understanding that the experience and individual response to breathing symptoms may, in part, be subjective, it should be recognized that socio-cultural influences may impact variations in how these symptoms are reported by individuals with PASC. A detailed review of breathlessness as it applies to

TABLE 3 Health Equity considerations and examples in post-acute sequelae of SARS-CoV-2 infection (PASC): Breathing Discomfort

Category	Comment	What is Known	Clinical Considerations
Disability *Example: People with spinal cord injuries (SCI)*	Individuals with SCI are medically complex and should be treated by clinicians who have experience with this population.	Individuals with SCI often have decreased lung capacity with pre-existing breathing issues that may be due to paralysis of respiratory muscles and/or deconditioning.[73] Sleep disordered breathing is also significantly higher in people with SCI than the general population.[74] Autonomic dysfunction is also a well-described issue. A systematic review in SCI patients infected with COVID-19 suggested that intensive care unit admissions were lower, and mortality may be higher than expected—a concerning issue that deserves further attention.[75] People with SCI are known to have numerous co-morbid conditions that pre-dispose them to worse outcomes (eg, obesity, diabetes, cardiac disease). For this population, the pandemic has resulted in less access to rehabilitation services including physical therapy, reduced access to community resources with adaptive exercise equipment and opportunities, and increased social isolation with deleterious effects on mental health.[76]	In the aforementioned systematic review, approximately one-third of patients with SCI and acute COVID-19 infection presented with dyspnea.[75] The literature is sparse regarding PASC symptoms, diagnosis, and treatment in the SCI population. Clinicians should be aware that people with higher levels of SCI (eg, cervical spinal cord) are at greater risk for respiratory compromise. Sleep-related breathing disorders should be given a high priority for diagnostic investigation. Pneumonia and other types of infections may present subtly with initial symptoms of increased spasticity. Reports providing some exercise and physical activity guidance during the pandemic for individuals with SCI have been published, and though these do not specifically address PASC, there may be some information that is helpful clinically. For example, one report suggests that individuals within SCI support groups could remind each other to engage their muscles (eg, using resistance bands) for at least 1 minute every hour.[76] Installing new adaptive equipment may be helpful and improve safety (eg, grab bars). Some individuals might have difficulty managing supplemental oxygen (ie, oxygen tubing, portable oxygen) and require adaptations or assistance. Protocols for future research that will assist with breathing-related clinical recommendations in this population are underway.[77]
Insurance *Example: Individuals who are uninsured, underinsured, or cannot afford access to recommended healthcare services*	Insurance coverage, or lack thereof, should be considered when devising a treatment plan addressing breathing-related issues in PASC. Encouraging patient engagement and addressing psychosocial factors may improve adherence with treatment recommendations.[78]	People with mild to severe COVID-19 infections may experience PASC-related breathing issues. After COVID-19 infection, the predictive risk factors of pulmonary fibrosis are advanced age, smoking, chronic alcoholism, illness severity, mechanical ventilation and length of intensive care unit stay.[79] Persons with severe fibrotic lung disease	Consider use of the lowest cost diagnostic testing such as chest radiograph (X ray), computed tomography (CT) of the chest, pulmonary function tests (PFTs) and treatment interventions. Some individuals may have difficulty affording generic or name brand prescription oral medication, metered dose inhalers, nebulizers, supplemental oxygen, continuous positive airway pressure (CPAP), and/or bi-level

(Continues)

TABLE 3 (Continued)

Category	Comment	What is Known	Clinical Considerations
		after COVID-19 may require lung transplantation.[80]	positive airway pressure (Bi-PAP) machines to assist with breathing. Long-term pulmonary follow-up may be required,[81] leading to higher healthcare costs. Referral to social services or community groups may assist persons with finding local support.
Racial / Ethnic Minority Groups *Example: People who identify as Black (including African-American), American-Indian/Alaska Native, Pacific Islander, Asian-American, and Mixed Race, and/or Latino/Hispanic (ethnicity)*	There are many reasons why the COVID-19 pandemic has a disparate impact on certain racial and ethnic minority groups, with higher rates of death in African-American, Latino/Hispanic and Native American communities.[82]	People who identify with minority groups may be at higher risk for COVID-19 infection, severe disease, and mortality. For example, one retrospective study of acute COVID-19 patients found that Hispanic individuals had a 2.76 odds ratio of decompensation and transfer to the intensive care unit within 24 hours of admission.[83] As COVID-19 disproportionately affects the case count and severity of disease in Hispanic patients, this cohort may require more respiratory related rehabilitation after hospitalization. In comparison, a cross-sectional study showed that amid patients diagnosed with COVID-19, both Black race and poverty were associated with higher risk of hospitalization, but only poverty was associated with higher risk of intensive care unit admission.[84] Similarly, non-Hispanic American Indian and Alaska Native (AI/AN) persons account for 0.7% of the United States population, yet 1.3% of COVID-19 cases.[85] Not only diagnosis but assessment and treatment related to breathing can be influenced by race and ethnicity. In two large cohorts, Black patients had nearly three times the frequency of occult hypoxemia that was not detected by pulse oximetry as White patients.[31] Although a discrepancy between measurements of pulse oximetry and arterial blood gas oxygen saturation was not found for all Black patients, unadjusted analysis found a discordance in 17% of Black patients and 6.2% in White patients.[31]	Racial and ethnic minority groups are more exposed to living and working conditions that predispose them to worse outcomes. Underpinning these disparities are long-standing structural and societal factors that the COVID-19 pandemic has exposed. As COVID-19 disproportionately affects the case count and severity of disease in minority groups, these individuals may require more respiratory related rehabilitation after hospitalization. Additional attention should be placed on the clinical symptoms of individuals with melanated skin tones, as the dependence on pulse oximetry alone to triage patients and adjust supplemental oxygen levels may place patients with darker skin tones at increased risk for hypoxemia.[30] For abnormal pulse oximeter readings, clinicians may consider assessing appropriate temperature of the area being tested, repeating pulse oximetry on a different extremity, encouraging deep breathing for adequate ventilation when possible or obtaining an arterial blood gas measurement. Patient clinical status and medical risk should guide individualized clinical decision making with consideration of advanced pulmonary workup when indicated.

(Continues)

TABLE 3 (Continued)

Category	Comment	What is Known	Clinical Considerations
Age *Example: Younger individuals and older individuals*	Many clinical trials have gaps in the inclusion of people across the age continuum, particularly children and older individuals.	*Younger*: Breathing problems may occur in children due to underlying conditions such as asthma and this may be exacerbated by acute COVID-19 infection. The American Academy of Pediatrics has made general recommendations regarding return to play for young athletes, and these are generally based on expert opinion with much of the advice focused on when to refer to cardiology or get cardiac testing.[86] Children infected with acute COVID-19 are susceptible to Multisystem Inflammatory Syndrome in Children (MIS-C). Reports have found that some minority groups, such as Black and Hispanic children are disproportionately affected.[87] Depending on clinical cardiopulmonary symptoms related to MIS-C, some children will require a gradual return to usual activities and play.[88] *Older*: Patients in older age categories often have worse outcomes. For example, one report notes that individuals over age 55 have had increased hospitalization, delayed clinical recovery, increased pulmonary involvement, and faster disease progression.[89]	*Younger*: Clinicians should be prepared to provide medical clearance, complete school forms, pre-participation sports physical examinations, and provide additional prescriptions for medication when needed (eg, rescue inhalers - one to be kept at school, one to be kept at home, one to be kept on their person if maturity level allows). *Older*: Elderly individuals may need long term services and support after COVID-19 infection. Responsibility for this care can fall to unpaid and untrained family members at home.[90] Coordination of safe return to community care after hospitalization in the elderly population is recommended.
Environmental exposure *Example: People who live in low socioeconomic environments that expose them to various types of environment-related stressors*	Social determinants of health play a role in healthcare outcomes,[91] especially in disorders of breathing. Public health measures and health policy legislation must continue to address these factors.[92]	Cough can persist for weeks or months after COVID-19 infection, often accompanied by chronic fatigue, cognitive impairment, dyspnea,[93] or pain. Elevated levels of air pollution have shown a positive association with COVID-19 mortality rates after considering area-level confounders.[94] Air pollution may also have a negative impact on the immune system as well as exacerbations of asthma and chronic obstructive pulmonary disease.[95]	Difficulty breathing during PASC will have an impact on many areas of daily life, more compounded by the need to manage supplemental oxygen. Neighborhood factors such as the need to walk, ride a bicycle or use carpool/ride-share as means of community mobility expose additional barriers during recovery. Some individuals rely on mass transportation within the community to travel to school, work, and leisure activities. Self-care tasks requiring moderate to high physical exertion may need modification (eg, showering and dressing lifting and caring for children or pets, taking clothes to laundromat for laundering). Access to formalized physical therapy, occupational

(Continues)

TABLE 3 (Continued)

Category	Comment	What is Known	Clinical Considerations
			therapy and pulmonary rehabilitation may be limited due to proximity to rehabilitation facilities. Economic factors such as ability to take time off from work/school for medical evaluations and treatments/ rehabilitation will also impact recovery and should be taken into consideration when developing a patient-centered comprehensive treatment plan.
Biologic sex *Example: Pregnant women*	Sex differences should be considered for both the diagnosis and treatment of PASC-related breathing issues.	Pregnant women frequently have pregnancy-related breathing issues, especially during the last trimester. Deep vein thrombosis and pulmonary embolism are well documented complications in pregnancy, and women with some underlying conditions such as sickle cell disease, may be particularly at risk for pulmonary complications.[96] Pregnant women may also be at higher risk for more severe COVID-19 infections and symptoms, particularly if they have comorbidities and other characteristics (eg, older age, diabetes, obesity).[97] During the acute infection, risk assessment is critical regarding the use of medications, oxygen therapy, and other interventions as these may impact maternal and fetal outcomes.[98]	Pregnant women who have had COVID-19 infections may experience pregnancy-related breathing issues and these may be exacerbated by PASC-related breathing problems. The differential diagnosis for breathing problems should include pulmonary embolism. Pregnant women often require alternatives to diagnostic testing (eg, to avoid radiation exposure). Safety for the patient and fetus is a primary concern, and the risks and benefits of medications and other treatment interventions should be assessed by clinicians. Exercise prescriptions may be impacted by both pregnancy-related symptoms (eg, nausea/vomiting, low back pain, pre-eclampsia) and PASC-related symptoms (eg, dyspnea, hypoxia).

Note: This table is included to provide additional information for clinicians who are treating patients for PASC-related breathing discomfort and respiratory sequelae. This is not intended to be a comprehensive list, but rather to provide clinical examples as they relate to health equity, health disparities, and social determinants of health. The literature demonstrates that all marginalized groups face socioeconomic barriers and access to care barriers, though these may or may not be barriers for a specific individual patient. People with intersectional identities (eg, those who identify with more than one underrepresented or marginalized group), often face enhanced levels of bias and discrimination.

historical, cultural, and existential relationships is beyond the scope of this report; however, it is important to note that these may play a role in the clinical evaluation.[29]

At the time of office evaluation, physical examination should include a detailed cardiac and pulmonary exam. Vital sign measurement should include pulse oximetry measurement at rest and with ambulation around the office or the individual's current environment if the visit is conducted virtually and this testing is feasible. This can serve as a simple and widely available screen to guide clinicians when prioritizing the next steps in diagnostic evaluation.

Overnight oximetry testing may also be considered for individuals with documented exertional or resting hypoxemia or those with daytime supplemental oxygen requirements, to determine overnight oxygen requirement. Further details regarding sleep disorders and evaluation will be addressed in forthcoming PASC Consensus Guidance Statements.

Vulnerable populations

It is notable that clinicians should be aware that patients with darker skin pigmentation may have pulse oximetry readings that are falsely higher than their actual arterial oxygen saturation, and thus experience occult hypoxemia[30]—previously described as oxygen saturation greater than 92% on pulse oximetry yet less than 88% on concurrent arterial blood testing.[31] In a recent retrospective analysis of hospitalized adults who underwent arterial blood gas measurements, Black patients had nearly three times the frequency of occult hypoxemia (11.4%) not detected by pulse oximetry, compared to White patients (3.6%).[31] Therefore, if mild hypoxemia accompanies dyspnea in patients with darker skin pigmentation, clinicians should consider repeat testing with an alternative oximeter, ambulatory oximetry, and potentially arterial blood gas if there is concern for occult hypoxemia. Pregnant patients may also have unique explanations for breathing discomfort and testing considerations that differ from those for non-pregnant patients (Table 3: Health Equity Considerations and Examples in Patients with PASC: Breathing Discomfort). Prompt evaluation and workup of new shortness of breath during pregnancy should be considered.

Characterizing breathing discomfort

A systematic characterization of breathing discomfort can also help to direct the assessment and treatment plan. Standard measures of breathing discomfort, such as the Multidimensional Dyspnea Profile (MDP), include a simple quantification of the breathing discomfort on a scale of 0 to 10, with 10 being the most severe discomfort. The trajectory of dyspnea is important, and repeated administration of a scale can help to quantify how the severity has changed over the post-acute period. These measures also allow for the characterization of the quality of breathing discomfort, including how it impacts activity level (eg, Modified Medical Research Council Dyspnea Scale) or what the sensation feels like (eg, MDP). Example measures of symptoms and activity related to breathing discomfort include:

1. Multidimensional dyspnea profile assesses sensation and severity of dyspnea sensation[32]
2. Modified Borg Dyspnea Scale is a means of quantifying the severity of the dyspnea sensation
3. Modified Medical Research Council dyspnea scale assesses impact of dyspnea on physical activity
4. Duke Activity Status Index assesses limitations in physical activity[33]

Testing

Pulmonary function

For patients with persistent symptoms that are not improving ∼8 weeks after symptom onset, or are worsening prior to that time, further evaluation with PFTs should be considered. This serves as a screen for pulmonary disease that may require additional evaluations in consultation with a pulmonologist and may direct specific treatment approaches. Although the exact timing of testing may be individualized, it is our experience that waiting ∼8 weeks provides adequate recovery from the acute phase of COVID-19 and reduces the overuse of testing in individuals who would otherwise recover within this time. Given that some degree of breathing discomfort is common in the early post-acute period of COVID-19,[5] a period of observation of the trajectory of recovery, prior to extensive testing, may be appropriate when history and examination is otherwise reassuring.

Where available, PFTs performed in a dedicated pulmonary function lab are recommended, providing comprehensive and accurate testing, including spirometry, lung volumes, and diffusion capacity. In settings where this is not available, in-office spirometry can be obtained. As discussed later, interpretation and further evaluation of abnormal PFTs can be undertaken in consultation with a pulmonologist. The need for repeat or serial assessment if patients have worsening symptoms can also be considered in consultation with a pulmonologist.

Imaging

Chest imaging is an important additional step for patients with abnormal oxygen saturation, persistent productive or dry cough, abnormal pulmonary exam, and/or impairment on pulmonary function testing. A chest radiograph should be considered as an initial step when performing imaging for most patients. However, if a chest radiograph does not explain the aforementioned abnormalities, in the presence of ongoing signs or testing abnormalities concerning for pulmonary disease, a chest CT scan should be considered to exclude or further characterize a possible pulmonary disorder. This chest CT can be performed as a high-resolution non-contrast scan in most patients, whereas a contrast-enhanced study is indicated if there is clinical concern for acute pulmonary embolism. In addition, a high-resolution non-contrast chest CT should be considered when abnormalities are identified on initial chest radiograph, to better define these abnormalities. Finally, clinicians can consider ordering a chest radiograph in settings where respiratory symptoms are not accompanied by other objective findings yet remain otherwise unexplained. Decisions regarding chest imaging may also be guided by a dedicated PASC clinic or a pulmonologist familiar with PASC, where available.

If patients have previous chest imaging available from the acute phase of illness, this should be obtained when possible, for comparison to follow-up imaging. Clinicians should be aware that residual imaging findings following viral pneumonia may persist beyond 4 to 6 weeks,[34] and the trajectory of findings along with clinical trajectory are important in this setting. Although follow-up imaging may be valuable in the setting of persistent or new symptoms, routine follow up chest imaging for viral pneumonia is of limited value in the absence of persistent symptoms or other clinical concern.[35]

Cardiac evaluation

Among patients who do not have evidence of lung disease, cardiac causes of dyspnea should also be considered. Cardiac testing is of limited value in many patients with breathing discomfort who do not have a history or physical examination suggestive of a cardiac cause and is best considered on an individual, rather than routine, basis. Assessment and treatment of cardiovascular complications of COVID-19 will be addressed in a subsequent consensus statement by our Collaborative. Clinicians may consider the best approach to testing in consultation with a PASC clinic or cardiologist, where available. Breathing discomfort may be associated with chest pain that can be similar to chest pain reported with coronary artery disease in some patients with PASC. For these patients, clinicians should direct testing to exclude cardiac pathology on an individual basis prior to considering rehabilitation approaches discussed below.

Breathing discomfort can be severe and disabling, regardless of whether an etiology can be identified on testing or not. Emotional distress associated with breathing discomfort has been noted, and a supportive and empathic approach by clinicians is essential to achieve functional improvement from these disabling symptoms. Additional contributors and exacerbating factors for dyspnea include muscular deconditioning following acute illness, anxiety, depression, post-traumatic stress disorder, medication effects, anemia, and dysautonomia.

Assessment in rehabilitation setting

Patients who are referred for rehabilitation therapies should be evaluated with objective measures of activity performance. These measures should be individualized to the patient's abilities, including consideration of modifications necessary to accommodate neurological impairment, fatigue with post-exertional malaise, dysautonomia, or other limitations. During a clinician's initial office evaluation, consider in-office measures such as 30-second sit-to-stand and 2-minute step tests.[36-41] In a physical therapy setting, more extensive measures such as 6-minute walk test may be performed. Alternative tests exist for patients requiring more in-depth assessment.

TREATMENT RECOMMENDATIONS DISCUSSION

Patients with abnormal PFTs or pulmonary exam or new oxygen desaturation with exertion, should be further evaluated by a pulmonologist

Individuals with PASC may present with a spectrum of pulmonary sequelae, ranging from shortness of breath and breathing discomfort with or without objective abnormalities on pulmonary testing, to organizing pneumonia, pulmonary fibrosis, and venous thromboembolism.[17-19] Patients with more severe initial COVID-19 more often manifest objective abnormalities on pulmonary testing.[22] Similar to survivors of ARDS, dyspnea is among the most common symptom of PASC, and patients may have objective abnormalities in cardiopulmonary health and fitness, including a restrictive pattern and impaired diffusion capacity on pulmonary function testing; decreased 6-minute walk distance, with or without hypoxemia; decreased peak oxygen consumption on cardiopulmonary exercise testing; and persistent ground-glass opacities or fibrotic changes on cross-sectional thoracic imaging.[16-18]

Following initial evaluation, as discussed in the assessment recommendations, patients with objective resting or exertional hypoxemia, abnormal PFTs, and/or abnormal imaging should be evaluated by a pulmonologist to guide additional assessment and treatment. Pulmonology evaluation should also be considered if abnormal breath sounds (such as rales, wheeze, or rhonchi) remain persistent, unexplained, or unresolved after management in the primary care setting.

Although progressive inflammatory or fibrosing lung disease following COVID-19 appears uncommon, the long-term trajectory of lung disease for those patients who do develop these changes is not yet known. Some patients may also present with organizing pneumonia necessitating treatment with oral glucocorticoids.[23] Thus patients with objective abnormalities in pulmonary testing or chest imaging warrant further evaluation with a pulmonologist.

Use of oxygen therapy

The use of long-term oxygen therapy in patients with hypoxemia is extrapolated from evidence that supplemental oxygen improves survival in patients with chronic obstructive pulmonary disease with severe hypoxemia.[42,43] Symptom mitigation is also significant with correction of resting or activity-induced hypoxemia with appropriate oxygen delivery. Thus in patients with any form of lung disease that results in chronic hypoxemia, oxygen therapy should be provided[44] according to established criteria. These criteria refer to measures at sea level, and may be modified for local conditions such as high-altitude settings:

Group I—patients with significant hypoxemia evidenced by any of the following:

- An arterial PO_2 at or below 55 mm Hg, or an arterial oxygen saturation at or below 88%, taken at rest, breathing ambient air.
- An arterial PO_2 at or below 55 mm Hg, or an arterial oxygen saturation at or below 88%, taken during sleep for a patient who demonstrates an arterial PO_2 at or above 56 mm Hg, or an arterial oxygen saturation at or above 89%, while awake, or a greater than normal fall in oxygen level during sleep (a decrease in arterial PO_2 more than 10 mm Hg, or decrease in arterial oxygen saturation more than 5%) associated with symptoms or signs reasonably attributable to hypoxemia (eg, impairment of cognitive processes and nocturnal restlessness or insomnia).
- An arterial PO_2 at or below 55 mm Hg or an arterial oxygen saturation at or below 88%, taken during exercise for a patient who demonstrates an arterial PO_2 at or above 56 mm Hg, or an arterial oxygen saturation at or above 89%, during the day while at rest.

Group II—patients whose arterial PO_2 is 56 to 59 mm Hg or whose arterial blood oxygen saturation is 89% or higher, if there is evidence of:

- Dependent edema suggesting congestive heart failure, or
- Pulmonary hypertension or cor pulmonale, determined by measurement of pulmonary artery pressure, gated blood pool scan, echocardiogram, or "P" pulmonale on electrocardiography (ECG) (P wave greater than 3 mm in standard leads II, III, or AVFL), or
- Erythrocythemia with a hematocrit greater than 56%.

We recommend using similar guidelines for the prescription of supplemental oxygen therapy for individuals with PASC who have rest, activity, or nocturnal hypoxemia.

During activity and rehabilitation, oxygen flow rate should be titrated to facilitate rehabilitation therapy and maintain oxygen saturations >90%. As mentioned earlier, patients using home pulse oximeters should be advised to obtain U.S. Food and Drug Administration (FDA)–approved devices for optimal accuracy.[45] Although adequate oxygenation can be achieved at low flow rates using nasal cannulas, if higher flow rates are needed, a face mask should be considered. Supplemental oxygen can mitigate symptoms and enhance exercise capacity in individuals with PASC and hypoxemia.[46] Portable tanks or portable oxygen concentrators can optimize community mobility, ability to participate in rehabilitation, and quality of life. Large tanks, liquid oxygen, and wall plug-in concentrators are acceptable for home use. Humidification should be considered for home O_2 delivery systems.

Education on the optimal use of oxygen is best done in a setting of pulmonary rehabilitation or in a supervised exercise setting. Self-monitoring of oxygen saturations with FDA-approved pulse oximeters and titration of oxygen flow can also be taught to individuals with PASC who require oxygen therapy to allow for better self-monitoring and appropriate use of oxygen in a home or rehabilitation setting.

Rehabilitation therapies

Rehabilitation therapies for individuals with PASC who are functionally limited by respiratory sequelae can address dyspnea and breathing abnormalities, fatigue, balance impairments, peripheral and pulmonary muscle strength, endurance, gait, and promote energy conservation and facilitate return to activity. In individuals with PASC and lung sequelae who also have accompanying fatigue and/or dysautonomia, the most physically limiting factor should be considered when deciding whether these approaches to rehabilitation are appropriate. For example, if fatigue is the most physically limiting symptom, clinicians should refer to previously published guidance statement regarding the assessment and treatment of fatigue.[15]

The use of simple functional measurements of capacity, as outlined previously, can allow for monitoring of functional status and measuring improvement over time in response to the rehabilitation therapies. Here, we outline rehabilitation approaches for multiple

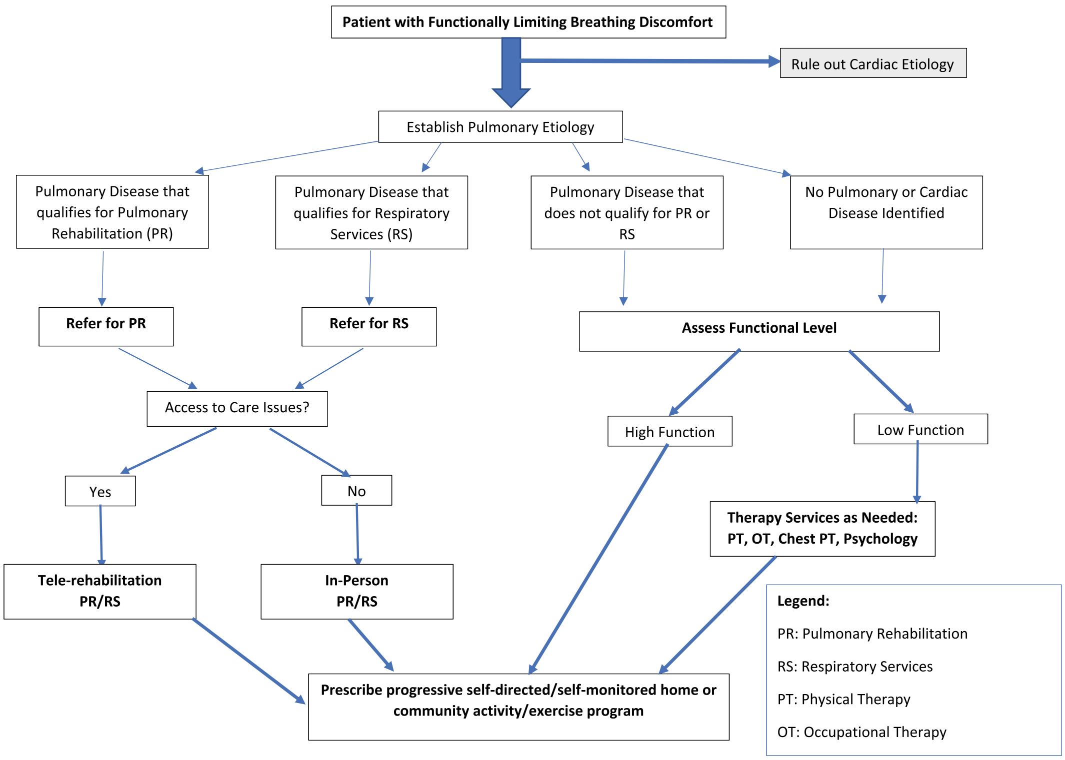

FIGURE 1 Decision tree: rehabilitation approaches for breathing discomfort

potential patient populations, including patients with breathing discomfort with and without accompanying impairment in pulmonary function. Figure 1 provides "high-level" considerations for these rehabilitation options, based on the initial assessment parameters. The eventual goal for rehabilitation management is an independent self-monitored home or community-based activity or exercise program.

Pulmonary rehabilitation and respiratory services

Individuals with PASC who have persistent respiratory symptoms and are functionally limited and have abnormal PTFs that meet criteria for pulmonary rehabilitation or respiratory services should be referred.

The benefit of pulmonary rehabilitation and respiratory services for patients with chronic lung conditions is well established,[47] and it appears from early studies that there is also benefit in individuals with PASC with lung disease.[48-51]

Insurance coverage delineates between pulmonary rehabilitation and respiratory services. Pulmonary Rehabilitation for obstructive lung diseases is covered for individuals with Medicare and Medicaid medical insurance by the Centers for Medicare & Medicaid Services (CMS) and billed under the G0424 code. For individuals with Medicare and Medicaid who have restrictive lung disorders, CMS covers Respiratory Services under G237, G238 or G239 codes. Most non-Medicare and Medicaid insurance plans cover Respiratory Services for both obstructive and restrictive lung disorders. Pulmonary rehabilitation programs also provide respiratory services and determine the appropriate billing code based on diagnosis and insurance coverage.[52]

Pulmonary rehabilitation/respiratory services are multi-disciplinary programs of care for patients with chronic respiratory impairment.[53] Programs are individually tailored and designed to optimize physical and social performance and autonomy. They are an evidence-based, multidisciplinary, and comprehensive intervention for patients with chronic respiratory diseases who are symptomatic and functionally limited.

Given that individuals with PASC may develop interstitial lung disease (ILD) and a restrictive lung disorder, these patients may benefit from pulmonary rehabilitation/respiratory services. A recent Cochrane review[54] concluded that "pulmonary rehabilitation probably improves exercise capacity, symptoms, and quality of life, and can be performed safely in people with ILD, including those with idiopathic pulmonary fibrosis (IPF)."

Pulmonary rehabilitation/respiratory services are designed to reduce symptoms, optimize functional status, increase activity participation, and reduce health care costs by stabilizing or reversing manifestations of lung disease. Programs focus on monitored progressive aerobic exercise and include strength, posture, and breathing exercises, 2 to 3 times a week for 2 to 3 months. Oxygen management, energy conservation and pacing, medication management, nutritional guidance, emotional support, and education are also key components. Prior to starting pulmonary rehabilitation/respiratory services in individuals with PASC-related respiratory disease, consideration of co-existing cardiac disease is indicated. Significant cardiac disease may be screened for by history, physical examination, and basic testing, including ECG. In individuals with pre-existing cardiac disease or who are considered to be higher risk for co-existing cardiac disease, cardiac stress testing can be considered with exercise stress echocardiography or pharmacologic stress echocardiography. Consideration should be given to consultation with a cardiologist if significant cardiac disease is noted prior to pulmonary rehabilitation participation. For all individuals, a functional exercise stress test may be a requirement prior to enrollment at some programs. A 6-minute walk test prior to starting the program is routine. Ideally and if available, a pre-program cardiopulmonary exercise test (CPET) may identify comorbid conditions that can impact safety for exercise, quantify an initial exercise intensity, and be used in conjunction with a repeat CPET at program completion to chart progress and recalibration of ongoing exercise intensity. Although this testing may not be routinely recommended for all patients with PASC, as discussed previously, clinicians should be aware that some form of cardiopulmonary and functional capacity testing may be required for enrollment in pulmonary rehabilitation.

Access to formal pulmonary rehabilitation/respiratory services programs is not universal and there are known disparities in access to this care for individuals who identify with racial and ethnic minority groups.[55,56] Challenges limiting participation may include insurance coverage; program accessibility/availability; lack of provider and patient knowledge as to the indications and benefit of pulmonary rehabilitation; and racial, ethnic, gender and socioeconomic disparities. The COVID-19 pandemic has also resulted in limited availability of in-person services, temporary/permanent closure of some Pulmonary Rehabilitation centers, or limited enrollment numbers due to COVID public health measures.

Alternative delivery models are a potential solution. Tele-rehabilitation and programs of remote video rehabilitation can be considered. The public health emergency has seen the easing of restrictions on telemedicine and allowed for the expansion of remote therapy services. Future access for these services is unclear; the current status of telemedicine requirements is available from the Federation of State Medical Boards[57] and other government agencies such as the CMS.[58] Although more data are required, preliminary studies indicate that tele-rehabilitation for pulmonary disease can be more effective than no rehabilitation and may be as efficacious as face-to-face programs.[59] Self-directed mobilization and exercise programs guided with at-home manuals for patients where tele-rehabilitation or in person rehabilitation are not available is another option. These self-directed programs can be monitored remotely and allow for patients to advance activity and intensity of the exercises as their recovery allows.[60]

Of note, for all exercise-recovery programs and services prescribed for individuals with PASC limited by respiratory disease, providers should delineate vital sign and oxygen saturation parameters to optimize safety and minimize over-fatigue and symptom exacerbation. Target oxygen saturation of 90% or above is ideal, and prescription should specify the amount and methods of oxygen delivery at rest and during exercise and activity. Parameters for high and low systolic and diastolic blood pressure values and peak and minimum heart rates have been established.[52] Patients should be careful to document recovery in the days after activity to ensure that fatigue with post-exertional symptom exacerbation is not a present or emerging PASC impairment.

Rehabilitation therapy services for patients not qualifying for pulmonary rehabilitation/respiratory services

Individuals with respiratory symptoms and/or disease with functional limitations who do not qualify for or do not have access to pulmonary rehabilitation/respiratory services can benefit from “traditional” rehabilitation programs including physical therapy (PT) and occupational therapy (OT) interventions focused on the impairments and functional deficits identified. These programs focus on recovery from debility, improving compensatory strategies, and activity progress as individuals can tolerate. For individuals with predominantly respiratory symptoms such as cough and sputum management, respiratory therapists or chest physical therapists may be engaged to provide specific respiratory care, as discussed in the following section, including secretion management, symptom management, oxygen use, and use of ventilatory support if needed.[61,62] In addition, respiratory services should include education about the disease process and treatments.

Functional field exercise testing is acceptable for individuals with PASC and respiratory symptoms considered at lower risk or who do not qualify for pulmonary rehabilitation/respiratory services. These tests include 2-minute step tests and 6- or 12-minute walk tests to assess capacity, assist in exercise prescription, and help quantify progress. Standards for field testing in individuals with lung disease are established by the European Respiratory Society (ERS) and American Thoracic Society (ATS)[63] and are available online with normative comparison values through the Official European Respiratory Society/American Thoracic Society Technical Standard: field walking tests in chronic respiratory disease available at: European Respiratory Society (ersjournals.com).

Patients at baseline high performance pre-COVID-19

Competitive and recreational athletes may achieve substantial functional recovery in the post-acute period yet continue to experience respiratory symptoms that limit their ability to perform at their high-performance pre-COVID-19 baseline. Although they may have had a reassuring medical workup, including normal laboratory and imaging findings, their persistent symptoms and inability to achieve previous activity levels may warrant a referral to an exercise specialist, such as an exercise or sports medicine physician, exercise physiologist, or physical therapist.

Prior to consideration for return to play and participation in sport, it is recommended that the athlete initially be able to resume basic activities of daily living and ambulate 500 meters without excessive fatigue or breathlessness, have a minimum of 10 days of relative rest following symptom onset, and remain symptom free for 7 days while off medications. A five-stage graduated return to play protocol has been proposed[64] under medical supervision with symptom monitoring, progressive duration of activity and intensity levels using percentage of maximal heart rate, and progression from simple to more complex movements in order to facilitate a safe return to sport without symptom exacerbation. CPET can accurately quantify aerobic capacity in endurance-based recreational and competitive athletes. Data from the CPET can be correlated with symptoms, utilized to design an appropriate training program, and can be repeated at appropriate intervals to track improvements. A cardiac risk stratification approach has also been proposed for athletes in competitive sports given potential for myocardial injury even after mild COVID-19.[65] Cardiovascular complications of COVID-19 are addressed in a forthcoming PASC Collaborative consensus document.

Routine performance-based testing may not reproduce familiar symptoms. Higher level sport-specific testing including determination of maximal oxygen consumption (VO_2 max) through CPET may be considered to:

1. diagnose or monitor for comorbid cardiopulmonary conditions that may affect exercise performance,
2. establish baseline fitness levels (aerobic capacity or VO_2 max),
3. design an appropriate training program,
4. evaluate progress after completion of the prescribed exercise program, and
5. evaluate hemodynamic response to prescribed exercise program to determine whether it is safe to resume or participate in competitive sports.

Exercise testing protocols[66] can provide guidance on individualizing a testing protocol to fit the activity demands of the returning athlete most appropriately.

Other adjuncts to rehabilitation: breathing exercises and chest physical therapy

Patients may benefit from breathing exercises that improve control of breathing, recovery from breathing discomfort, and perception of breathing discomfort. Several self-directed educational resources exist to guide breathing therapy, as well as in-person rehabilitation, or online programs:

Organization	Online Programs	Website
1. Stasis	Breathing Rehab Program, Coaching and Group sessions, On-Demand, Statis App	https://www.stasis.life/
2. Pulmonary Wellness Foundation	Online Rehab/ COVID Bootcamp	https://pulmonarywellness.org/
3. Beth Israel Deaconess Medical Center	Consciousness and Cognition Education	https://www.bidmc.org/research/research-by-department/anesthesia-critical-care-and-pain-medicine/research-centers/sadhguru-center/education
4. Johns Hopkins Medicine	Coronavirus Recovery: Breathing Exercises	https://www.hopkinsmedicine.org/health/conditions-and-diseases/coronavirus/coronavirus-recovery-breathing-exercises

The prescription of airway clearance techniques (ACTs) may be beneficial in individuals with PASC who have cough productive of copious daily sputum and/or where evidence of bronchiectasis or retained secretions exists.[67] The benefit of ACT needs to be considered carefully against the added treatment burden given the time needed to complete ACT, and the cost of ACT devices if being used. The clinical rationale and evidence for use must support the prescription of ACT. If little or no impact will be made on the individual's outcome, routine ACT may not be beneficial. Individuals with severe respiratory muscle weakness following critical or intensive care admission may benefit from inspiratory muscle training (IMT).[68] The potential to cause exacerbation of PASC symptoms in individuals with dysautonomia, fatigue, and/or post-exertional malaise should be considered when prescribing therapies such as ACT and IMT in the rehabilitation program.

Pharmacologic therapies

In our experience, patients with breathing discomfort are commonly prescribed inhaled therapies as an early empiric trial of treatment. These most often include inhaled short acting beta-agonists and inhaled corticosteroids. We have also commonly observed patients prescribed limited courses of oral glucocorticoids to treat breathing discomfort. Given the overall safety of short-duration use of inhaled therapies, an empiric trial of these therapies may be considered on an individual basis at a treating clinician's discretion. However, any empiric trial should be accompanied by close monitoring of symptom response, side effects, and discontinuation if therapies do not benefit patients. Objective assessment such as PFTs, as discussed in this report, may ultimately help guide the use of such therapies. Our experience has been that these therapies do not commonly result in an improvement in symptoms or pulmonary function, and may be accompanied by side effects, particularly from oral glucocorticoids, or among patients with dysautonomia using beta-agonists. Therefore, we do not routinely recommend the use of pharmacotherapies if clinical assessment and testing do not support their use.

It is important to note that use of pharmacologic agents and supplements varies across PASC clinics. Prescribing medications or supplements should be considered on a case-by-case basis, recognizing the limited scientific evidence. In addition, there should be consideration of the out-of-pocket cost of supplements, the risk of medication interactions, lack of regulation, and possible side effects.

Strengths and limitations

The strength of these guidance statements includes their development within a collaborative of clinicians and patients with extensive professional and personal experience with PASC. Given the limited evidence regarding clinical care, we organized a nationwide group of experts treating patients at PASC clinics around the United States to conduct a rigorous approach to consensus generation. Methods for this consensus process were developed and published[14] prior to our initial consensus statement focused on fatigue,[15] and these methods have been applied carefully throughout the process.

As discussed earlier, these statements are intended to reflect current practice in patient assessment, testing, and treatments. Given the rapidly evolving nature of our understanding of PASC, a limitation of these recommendations may be due to the rapid evolution of knowledge over time. It is important to note that the recommendations should not preclude clinical judgment and must be applied in the context of the specific patient, with adjustments for patient preferences, comorbidities, and other factors. Another limitation is that this document specifically addresses post-acute persistent breathing discomfort and respiratory sequelae in the outpatient setting and does not apply to the acute assessment or management of breathing discomfort in urgent or emergency care settings.

Health Equity Statement

The American Academy of Physical Medicine and Rehabilitation (AAPM&R) recognizes the need to support equitable access to rehabilitation care for individuals with post-acute sequelae of SARS CoV-2 infection (PASC). The AAPM&R states that equitable access to care includes: (1) timely and local patient access to multidisciplinary care; (2) addressing inequities in the US health care system that result in diminished access to sustained quality care because of structural racism or socioeconomic factors; and (3) strengthened safety-net care, including disability evaluation and benefits.[69]

Each of the AAPM&R's PASC guidance statements were produced by a diverse and multidisciplinary team of subject matter experts with patient input. Although an in-depth discussion of health equity issues is beyond the scope of the PASC guidance statements, each one highlights health equity concerns and refers readers to other publications and resources. The term "health equity" has many different definitions, and they generally focus on ensuring that every person is able to achieve the highest level of health and function. For example, the Centers for Disease Control and Prevention (CDC) defines health equity as the opportunity for people to fulfill their full health potential and states that people should not be disadvantaged from achieving their potential because of social position or other socially determined circumstances.[70] The Centers for Medicare & Medicaid Services (CMS) uses the definition established in Executive Order 13985, issued on January 25, 2021, that states that equity is "the consistent and systematic fair, just, and impartial treatment of all individuals, including individuals who belong to underserved communities who have been denied such treatment, such as Black, Latino, and Indigenous and Native American persons, Asian Americans and Pacific Islanders, and other persons of color; members of religious minorities; lesbian, gay, bisexual, transgender, and queer (LGBTQ+) persons; persons with disabilities; persons who live in rural areas; and persons otherwise adversely affected by persistent poverty or inequality."[71] There are many root causes of health disparities, some of which fall under the categories within social determinants of health (SDOHs). Examples include, but are not limited to, socioeconomic status, neighborhood, availability and access to healthy food, and access to a high-quality education.

In addition to advocating for equitable access to rehabilitation care for all persons with PASC, the AAPM&R supports four "Principles of Inclusion and Engagement," which include: (1) valuing diverse group composition (a diverse group is more representative of AAPM&R's membership and volunteers may be selected as a member of a particular community to enhance diversity of thought and experiences); (2) mutual respect (cultivating a receptive space for differing opinions and viewpoints); (3) talent and skill-based selection for leadership opportunities (ensuring that broad criteria of diversity of experience, talent and knowledge are incorporated and removing barriers to involvement that support an equitable environment); and (4) comprehensive collaboration (building community among various member constituent and bringing together different perspectives).[72] Readers of the PASC guidance statements are encouraged to consider the recommendations through the lens of health equity in order to improve access to rehabilitation care for all individuals with PASC.

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DISCLOSURES

Dr. George Alba has a patent pending for a novel antibody, unrelated to the present work, and owns stock in Vertex Pharmaceuticals. Dr. Matthew Bartels reports grants from the National Institutes of Health (NIH) and AposTherapy, unrelated to the present work. Dr Mathew Bartels is a consultant to EHE Medical, LIH Medical, and Neofect Corporation. Sarah Sampsel is a consultant paid by AAPM&R for writing and project management. Dr. Jamie Wood is a non-financial advisor to Statis, a company mentioned in the manuscript as an option for breathwork training. Unrelated to this work, Dr. Julie Silver participates in research funded by the Binational Scientific Foundation and is a Venture Partner at Third Culture Capital.

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